Beat Machine S3 – User Manual (beta 3)

Functions in Red still need implementing

Modes / Inner Button	Bank	Colour	Click Action	Hold Click	Combo Hold Click	Dial 1	Dial 2	Dial 3	Dial 4	Dial 5	Dial 6	Dial 7	Dial 8
Global	0		Outer buttons toggle step state.			**Filter (LP on Wooden, Dual on PCB)**	**Density**	**Evolve**	**Chaos**	**Drive**	**Tempo**	**Length**	**Volume on PCB, HPF on Wooden**
Mix **A**	1	Red		1-8: Play sound		Vol 1	Vol 2	Vol 3	Vol 4	Vol 5	Vol 6	Vol 7	Vol 8
FX **B**	2	Orange		1-16: Delay time		Reverb Level	Reverb Time	Delay Level	Delay Time	Accel X Amnt	Accel Y Amnt		LED brightness
Sequence **C**	3	Light Green	Tap Tempo	1-16: Select factory pattern preset	FX - 1-16: Select user pattern preset Mix - 1-16: Save user pattern preset Sound – 1-12: Diatonic transpose	Half or Double Time	Step Size	Slice Amnt	Swing	Accent	Tempo Fine Control	Key	Scale
Generate **D**	X	Blue	New Euclidean Kit Pattern		Mix – Two-three kit pattern FX – Probability Pattern Seq – Polyrhythm kit pattern Stutter – Init empty pattern Part – Euclidean part Part & Mix – Two-three part Part & FX – Probability Part Part & Seq – Polyrhythm part Sound – Generate Kit sound Part & Sound – Gen Part Sound								
Play **E**	X	Green	Start/Stop										
Stutter **F**	4	Purple		1-16: Select stutter step(s)	Generate – Init empty pattern Part – Stutter current part only	Stutter Length	Curr Part Pitch				Global LPF	Curr Part tone decay	Curr Part noise decay
Part **G**	5	Aqua		1-8: Select part 9-16: Mute Part Double click 9-16: Solo Part	Generate – Euclidean part Gen & Mix – Two-three part Gen & FX – Probability Part Gen & Seq – Polyrhythm part Gen & Sound – Gen Part Sound Stutter – Stutter current part	Part BPM (& Seq fine control)	Part Density	Part Evolve	Rotate	Part Octave	Part Range	Part Length	Pitched/Unpitched
Sound **H**	6	Pink		1-16: Select factory sound preset	FX - 1-16: Select user sound preset Mix - 1-16: Save user sound preset	Osc Balance	Pitch	Mod Rate	Mod Depth	Noise Cutoff	Attack	Tone Decay	Noise Decay

Quick Start Guide

Power on. Beat Machine (BM) wooden, use Switch next to USB port. BM PCB, connect to a powered USB cable and connect audio to headphones or mixer.

UI Orientation

There are two rings of **LEDs** and **button** pairs. 16 in an outer ring and 8 in an inner ring. The outer ring shows and edits the rhythmic pattern of the current part. The inner ring relates to the BMs operating mode, or state, and these LEDs pair with 8 inner buttons.

The inner ring of **buttons** are labelled A to F and trigger actions (e.g. play and stop) or change 'modes'. *Hold* buttons G to C to temporarily change mode or *double-click* to lock to a mode. The dial's change function in each mode.

Dials are numbered 1 to 8 in clockwise order, starting at 12 o'clock (sorry, no dial labelling on alpha version). Dial adjustment triggers a yellow LED indicator of dial value on the outer LED ring. When modes change, dials change function. They need to pass the previous value (the latch target) before the parameter changes. When dials are moved the nearest outer LED is a latching indicator, if light red the dial is higher that the target value, if light blue the dial is lower that the target, when light yellow the dial is active and parameters are being updated.

Instant Gratification

Press PLAY (Button E, green LED) to start and stop a sequence – On BM Wooden if there is LED activity but no sound, check the speaker-toggle switch on the back.

Dial F adjusts tempo, … (see matrix chart), dial 8 adjusts Volume on the BM PCB. On BM Wooden, the centre dial adjusts the volume.

It's a **generative** drum machine, so generate a new sequence by pressing GENERATE (button D, blue LED).

It's **probabilistic**, and dial 2 controls step probability, from 0% on the left (drop out steps) through 100% in the middle (play pattern normally), and 200% on the right (fill in step gaps).

Manual Pattern Edit

The outer ring of buttons is numbered 1 – 16. Buttons **toggle steps** on/off in the current part (kick drum is part 1, snare is part 2, hats are part 3, …). Double-clicking outer ring buttons will add two 32nd notes to that 16th note block. Clicking to turn off a step, will clear all notes in that 16th note block.

Change **parts** by holding the PART button (G) to enter Part mode and press an outer button 1 – 8. LEDs will redraw to show the current part's pattern. Active steps in each part use a different colour, and non-active steps are a dull aqua colour. If patterns are less than 16 steps, LEDs on non-used steps are off.

Change part **volume** – hold the MIX button (A) to go into Mix mode and use dials 1 – 8 to adjust the volume of parts (voices/tracks/instruments) 1 – 8.

Mute parts by holding the PART button (G) and press outer button 9 – 16. LEDs 9-16 will flash when their parts are muted. Press again to unmute. Hold PART and double-click 9-16 to solo a part.

Initialise an **empty pattern** by holding the EFFECTS button (B) and pressing the STUTTER button (F). This also resets the pattern length to 16 steps, if required. Now add your own steps to each part.

Generative Pattern Algorithms

Euclidean rhythms - generate a new Euclidean pattern by pressing the GENERATE button (D).

Polyrhythms – generate a new polyrhythm by holding the SEQUENCE button (C) and pressing GENERATE (D).

2 and 3 step rhythms - generate a new Small Duration Pattern by holding the MIX button (A) and pressing GENERATE (D).

Probabilistic rhythms – generate a new structured random pattern by holding the EFFECTS button (B) and pressing GENERATE (D).

Generate Algorithmic Patterns for Each Part

For a new Euclidean pattern on the current part only, hold the PART button (G) and press GENERATE (D). For a new 2-3 step part, hold the MIX button (A) and press GENERATE (D). For a new Polyrhythm on the current part, hold the SEQUENCE button (C) and press GENERATE (D).

Updating individual parts with generated rhythms, allows patterns to have a mix of rhythms generated by different algorithms and then edited manually to your liking.

Each of the 8 parts uses a different highlight **colour** for active steps on the outer ring of LEDs. The same colours are used on the inner ring of mode button LEDs. Starting with red on inner button and part 1, yellow next, and so on clockwise. Looking at the inner LEDs can be a reminder of which colour is used for which part. Be aware that the LEDs can show different shades of a ‘colour’ at different intensities.

Modes and Dials

The BM defaults to Global mode where the **dials** correspond to Filter, Density, Evolve, Swing, Drive, Tempo, Length, and Volume, which effect all parts.

The dials can have other functions when in a different mode (see the matrix). Inner buttons other than PLAY and GENERATE are used to change mode. To temporarily enter a different mode, hold down an inner ring button then turn a dial. To lock the BM in a mode, double click an inner button. To go back to Global mode, click any inner button.

Change The Sounds

It's a **multitimbral** synthesizer. All sounds are generated in real time. Each part uses the same synthesis engine, but has its own sound settings.

Select a factory **sound preset** by holding the SOUND button (H) and pressing outer buttons 1-16 (only some slots currently have a factory preset...).

Edit sounds by holding the SOUND button (H) to enter Sound mode. Then use dials 1-8 to change sound parameters. Dial 1 changes the balance between tone and noise, dial 2 changes the tone pitch, … see the matrix chart above. Double-clicking on the SOUND button (H) will lock the BM in Sound mode for extended sound design sessions. Click the SOUND button again to exit back to Global mode.

All parts use the same synthesis engine, but by convention (and a bit by design) part 1 is a kick drum, part 2 is a snare, part 3 is hi hat, part 4 is a clap, part 5 is a clave, part 6 is a tom, part 7 is wooden percussion, and part 8 is metallic percussion. A synthesis architecture diagram is shown on the Synth Controls page of this user guide.

Go Nuts

Parts can be of different **lengths**, allowing for **polymetric** patterns. Hold the PART button (G) and turn dial 7 to adjust the length of the current part. Go to a different part and repeat the process.

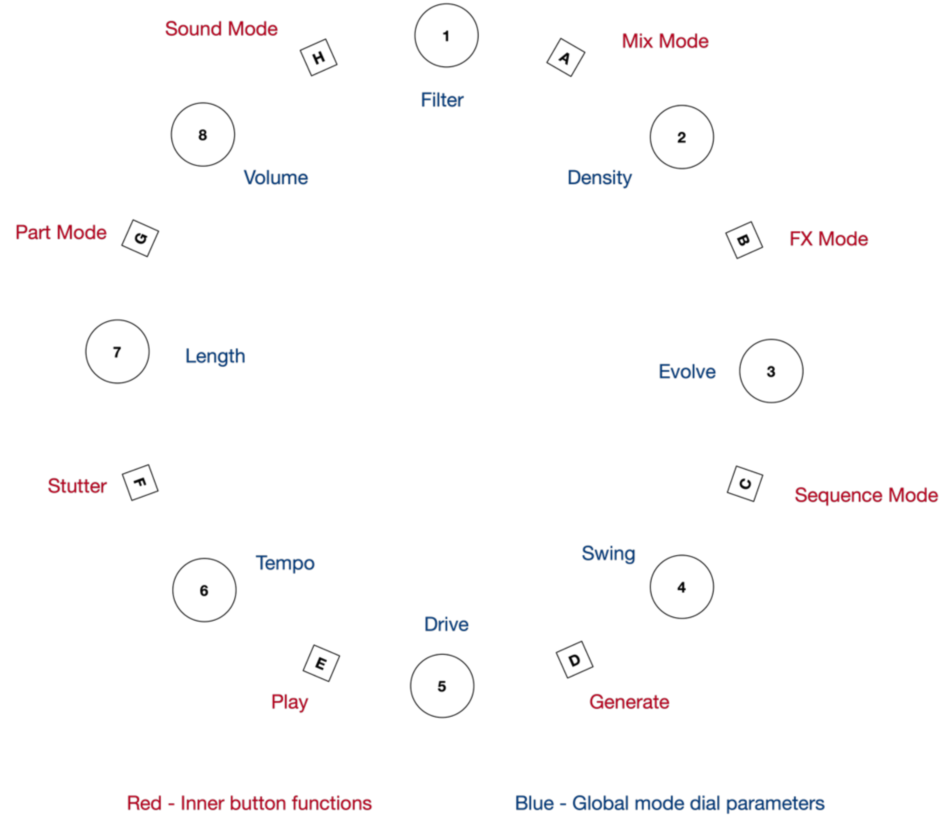
Rotate a part. Hold the PART button (G) and turn dial 4 to shift the metrical phase of the current part. (Think of Reich's 'Clapping Music').

Parts can be at different tempi, allowing for time **phasing** of patterns (Think of Reich's 'Piano Phase'). Hold the PART button (G) and turn dial 6 to change the tempo of the current part in 10 BPM increments. For finer tempo adjustments (1 BPM increments) Hold the SEQUENCE button (C) and turn dial 6. Change to another part and repeat for even more polymetric shenanigans.

Update Firmware

The latest Beat Machine firmware can be installed via a web browser, installation instructions and user manual can be accessed from the BM GitHub repository: https://github.com/algomusic/beatmachine

Global Mode Controls

Buttons	Dials	
• Play – Stop (E) • Stutter (F) • Part Select (G+1:8) • Part Mute (G+9:16) • Part Generate (G+D) • Pattern Generate (D) • Polymetric Gen (C+D) • 2&3 Step Gen (B+D) • Probabilistic Gen (A+D) • Init Pattern (H+D)	1. Filter 2. Density 3. Evolve 4. Swing 5. Drive 6. Tempo 7. Length 8. Volume	

Synthesis Controls

<table>
<tr><th>Buttons</th><th>Dials</th><th></th></tr>
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- Sound Mode (H)

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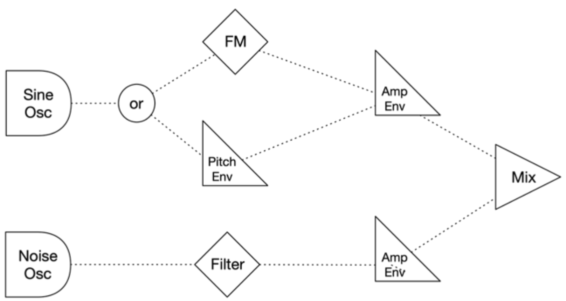
1. Oscillator Balance
2. Pitch
3. FM Ratio / Pitch Env On
4. FM Mod Depth / Pitch Env Depth
5. Noise Filter
6. Attack
7. Tone Decay
8. Noise Decay

Beat Machine S3 - Synthesis Architecture



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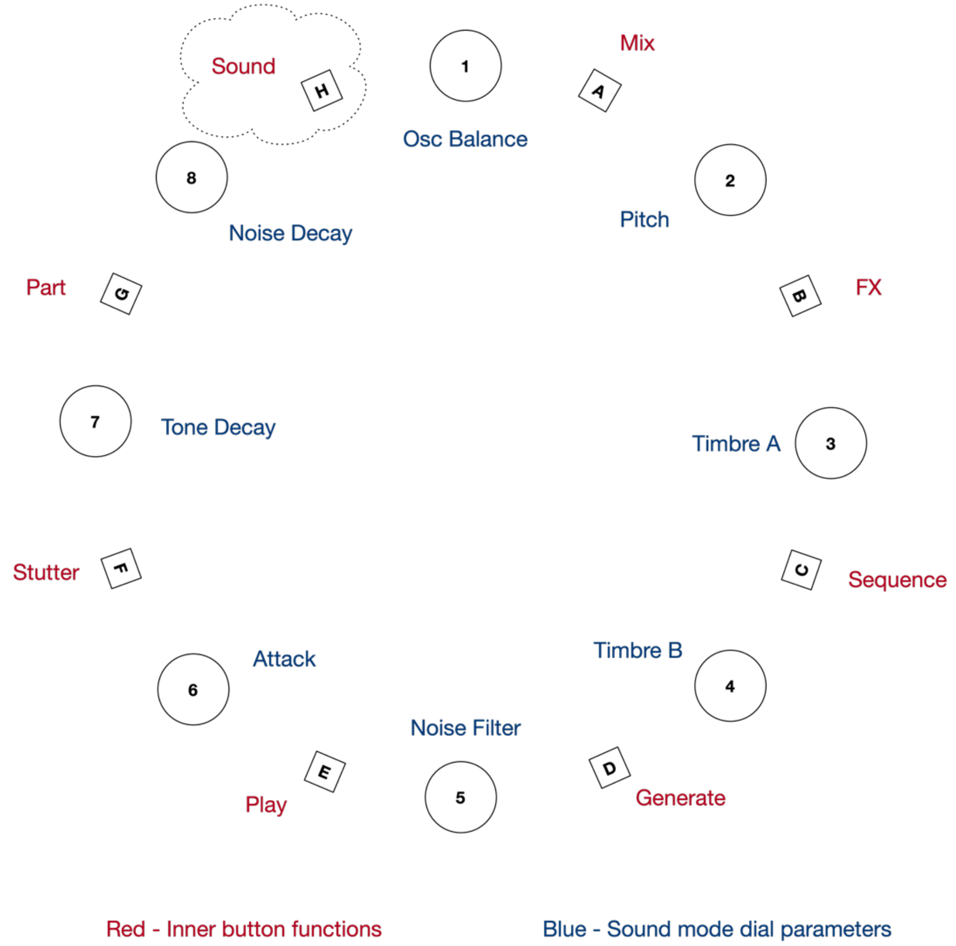

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Parameter details

Tempo – 120 BPM by default. When adjusting, BPM changes in 10 bpm increments at each of the 16 steps, from 40 BPM at step 1 to 120 BPM at step 9, to 190 BPM at step 16. To fine tune the tempo in 1 BPM increments, hold the SEQUENCE button (C) then turn dial 1 – this increases the base tempo up to 15 BPM, from 0 increase at step 1. Change the tempo of the current part individually by holding the PART button (G) and moving dial 6 for 10 BPM increments or hold SEQUENCE (C) & PART (G) to increase it in 1 BPM increments. Finally, tap tempo using the SEQUENCE button (C) – 4 or more taps are required to set a tempo.

Density – Determines the probability that a step with sound, or not. Density defaults to 100% and ranges from 0% to 200% with 100% being at step 9.

Timbre – Most of the sound synthesis controls are obvious from their names, however the two timbre controls have dual functionality. Timbre control A (Sound mode, dial 3) switches between FM Ratio and Pitch Env On/Off. When dial 3 is at zero there is no FM and a **pitch envelope** on the tone is enabled. Timbre control B (Sound mode, dial 4) then controls the **depth** of the pitch envelope; with 0 in the centre, to the left is increasing upward pitch bend, to the right is increasing downward pitch bend (typical of a TR-808 kick or Simmonds tom). When in Sound mode and dial 3 is above zero, this turns on FM and adjusts the Carrier:Modulator **ratio**. Dial 4 then adjusts the modulation **depth**.

TBC...

Factory Presets

There are 16 built-in preset patterns and 16 preset sounds. All presets include 8 parts each. Patterns and Sounds are independent, so you can mix and match as you like. Factory presets can't be overwritten so they are always available.

User Presets

There are slots for 16 user preset patterns and 16 user preset sounds. All presets include 8 parts each. These user presets can be overwritten so you can update them as often as you like – for example, to put a collection together for a performance. Sound presets include the part volumes. Pattern presets include the length, tempo and density settings at the time of saving.

Preset Type	**Button Combo**	**Normal**	**With Part Held**
Load **User Sound**	Mix + Sound + Step	All 8 parts	Current part only
Load **User Pattern**	Mix + Sequence + Step	All 8 parts	Current part only
Load **Factory Sound**	Sound + Step	All 8 parts	Current part only
Load **Factory Pattern**	Sequence + Step	All 8 parts	Current part only
Save **User Sound**	FX + Sound + Step	All 8 parts	Current part only
Save **User Pattern**	FX + Sequence + Step	All 8 parts	Current part only